ENR 3.3 AREA NAVIGATION ROUTES

1 RANDOM RNAV OPERATIONS ACROSS THE SOUTH ATLANTIC OCEANS

1.1 South Africa has the delegated responsibility of providing service over the high seas in those areas of airspace which have been identified as part of and included within the South African Oceanic FIR. These areas have variously been identified as the Atlantic Ocean Random Routing Area (AORRA). Operating within these areas facilitate freedom to flight plan and operate along random tracks according to operational requirements, thereby contributing to operational and economic efficiency and at the same time advancing the quest to reduce carbon emissions.

2 INTRODUCTION

2.1 The Purpose of this section is to advise users of the implementation and to publish the proposed procedures and associated dimensions of the Atlantic Ocean Random Routing Area (AORRA).

2.2 The users will benefit by being able to utilise the most convenient flight trajectory. Compliance with RNAV10 will ensure the navigation precision required for operations in the area of RNAV operations.

3 GENERAL CONDITIONS FOR PARTICIPATION IN THE AORRA

3.1 Only those aircraft certified for RNAV 10 operations would be permitted to operate within the AORRA. Authorised operators indicate compliance by entering "X" in field 10 of the ATS Flight Plan.

3.2 This requirement shall ensure the level of navigation accuracy required for RNAV operations.

4 DESCRIPTION OF AORRA AIRSPACE

4.1 AORRA is designated as the area with vertical limits FL290 and FL410 within the, Atlántico, Dakar, Comodoro Rivadavia, Ezeiza, Johannesburg Oceanic, Luanda Oceanic and Montevideo FIRs bounded by a line joining the following coordinates:

4.1.1 Defined Area of the AORRA

Commencing at	600000S	0150000E
A straight line to	273000S	0150000E

Then straight lines to each of the following co-ordinates:

	173000S	0111300E

	094000S	0112400E (ONTAR)
Then via a 120 NM arc centered on Luanda to		
	074800S	0113000E (OPAPO)
	052000S	0100000E
	053000S	0085000E
	041000S	0063500E
	000000S	0063500E
	000000S	0263400W
	085400S	0315600W
	115500S	0325300W
	153400S	0361800W
	183000S	0384500W
	194300S	0345500W
	264500S	0434500W
	340000S	0500000W
	340000S	0513320W
	364530S	0531147W
	582106S	0530000W
	600000S	0530000W
	600000S	0150000E

Flights operating within the AORRA shall enter and exit AORRA via the following gates:

Johannesburg Oceanic	IBNEV	184706.00S	0114005.00E
	NIBEK	225831.00S	0131254.00E
	NIGAM	263356.20S	0143710.00E
	OKTEL	280753.81S	0150000.00E
	UVGOD	290943.27S	0150000.00E
	ALDOV	303712.00S	0150000.00E
	BUXIR	320000.00S	0150000.00E
	OKDOG	330500.00S	0150000.00E
	ITMEK	341200.00S	0150000.00E
	ITLIK	351600.00S	0145957.00E
	NEPUM	321551.00S	0322743.00E
	EGVOL	340249.00S	0310146.00E

	EPNET	37 00 34.00S	023 59 04.00E
	GEPAB	37 00 41.00S	018 30 42.00E
	IMKAM	31 14 20.00S	015 00 31.00E
	ANTOM	29 50 00.00S	015 00 00.00E
	GEPEV	35 04 42.00S	029 59 12.00E
	ETBOR	37 00 05.00S	027 42 42.00E
	NERAX	36 01 07.00S	029 01 40.00E
	AVANA	37 00 00.00S	026 00 00.00E

Luanda	URAPI	09 49 07.00S	003 48 07.00W
	OSUKO	09 01 01.00S	001 36 04.00W
	GAPEL	08 17 06.00S	000 19 00.00E
	TERBA	04 47 09.00S	000 35 00.00E
	OPAPO	07 48 00.00S	011 30 00.00E
	ONTAR	09 40 00.00S	011 24 00.00E

Atlántico	CIDER	24 07 49.00S	040 16 23.00W
	EKALO	22 26 00.00S	038 08 48.00W
	GARUP	18 51 48.00S	037 40 24.00W
	PORGA	18 40 48.00S	038 14 36.00W
	POLVO	18 35 24.00S	038 31 12.00W

Montevideo	BIVEN	36 35 00.00S	053 05 10.00W

Ezeiza	GUXOR	37 22 30.00S	053 00 00.00W
	BISUL	43 31 22.00S	053 00 00.00W
Comodoro Rivadavia	EGLAS	48 00 00.00S	053 00 00.00W
	IRIRO	60 00 00.00S	053 00 00.00W

4.2 **Automatic Dependent Surveillance / Controller Pilot Data Link Communication (ADS/CPDLC)**

4.2.1 ADS/CPDLC will be utilised in AORRA airspace by suitably equipped service providers to provide an ATS service to aircraft able to take advantage of this form of communication. Operators are to note that in some sectors of the random routing airspace, ADS/CPDLC is the primary form of communication, with HF as secondary means of communication.

4.3 Required Navigation Performance (RNAV) 10 procedures for aircraft operations within the AORRA

4.3.1 Only those aircraft certified for RNAV10 operations shall operate within the AORRA.

4.3.2 No aircraft shall flight plan to operate in the AORRA airspace unless it is RNAV10 certified to operate in this airspace by the State of Registry or the State of operator, as the case may be, except in the following circumstances:

i) The aircraft being initially delivered to the State of Registry or the State of the operator;

ii) The aircraft is certified but experienced navigation degradation and is being flown back to base or to a maintenance facility for repairs;

iii) The aircraft is engaged on a humanitarian or mercy flight;

iv) State aircraft.

4.4 APPROVAL OF AIRWORTHINESS/OPERATIONS

4.4.1 RNAV10 Approval - The operating or intending to operate in AORRA airspace shall obtain RNAV10 approval from the State's registry or States operator as appropriate and which user complies with the following conditions:

4.4.2 The aircraft satisfies specifications of "Minimum aircraft system performance specifications" (MASPS) of the State's registry.

4.4.3 The aircraft is operated under the conditions indicated in the RNAV10 operational approval issued by the user's State.

4.5 FLIGHT PLANNING IN AORRA

4.5.1 When it is intended to operate an aircraft in AORRA airspace, RNAV10 compliance shall be indicated by placing an "R" in box 10 of the flight plan form.

4.5.2 Flight plans shall contain entry and exit points to the AORRA and the estimated time for every 5° of longitude.

4.5.3 Those operators operating under the circumstances stipulated in paragraph 4.3.2 shall insert STS/NON RNAV10, in field 18 of the ICAO FPL.

4.5.4 Aircraft meeting the navigation requirements may flight plan any preferred track between the entry and exit gates.

4.5.5 Additionally, a waypoint shall be included for each FIR crossing.

4.6 OPERATIONAL PROCEDURES BEFORE ENTERING AORRA AIRSPACE

4.6.1 Before entering AORRA airspace the pilot-in-command shall verify that the required equipment to fly within AORRA airspace is operating normally and verify with the greatest possible accuracy the position of the aircraft through external air navigation aids.

4.6.2 If any equipment is not operating normally, the pilot should notify ATC before entering the AORRA airspace.

4.7 OPERATIONAL PROCEDURES AFTER ENTERING THE AORRA AIRSPACE

4.7.1 General Procedures

4.7.2 If an aircraft cannot continue the flight in compliance with the ATC clearance issued and/or cannot maintain the precision required for the specified navigation performance in the airspace, ATC is to be advised immediately.

4.8 Position reporting shall be required at Entry/Exit gate.

As well as any other position required by ATC.

05E
10E
15E
00 E/W
05W
10W
15W
20W
25W
30W
35W
40W
45W
50W

4.9 ROUTES IMPLEMENTATION IN AORRA

4.9.1 In order to enhance access to and from that portion of the AORRA which falls under the responsibility of Johannesburg Oceanic Control, the following entry/exit gates to the AORRA as reflected in the table below, in addition to the already existing entry/exit gates, have been established on that part of the Johannesburg Oceanic FIR (FAJO) boundary adjoining the Windhoek FIR Continental airspace. The establishment of the additional entry/exit gates coincides with the establishment of fixed continental routes transiting the Windhoek FIR in order to facilitate access to and from the entry/exit gates, complimenting the current established route network.

	LATITUDE	LONGITUDE
NEVEP	202000.00S	0121404.44E
ETUDU	214000.00S	0124321.18E

ANTEP	240000.00S	0133604.00E
DULGO	250000.00S	0135908.00E

4.9.2 Operators are to note that other than in emergency access/exit to/from the AORRA are restricted to the published entry and exit gates.

***NOTE**: In addition to the existing route system, the following ATS route segment has been implemented in the Johannesburg Flight Information Region (FAJA):*

UT394 Bi-directional

POSITION	COMMENT	LATITUDE	LONGITUDE
AVUSA	*X UQ19	272955.1042S	0221330.0871E
UTANI	FIR/FIR	271346.73S	0210034.80E

** The X indicates the route or portion of the route which crosses another route at the position indicated.*

Note:

a) Additionally to the above route segment, the following route, UT396 has been implemented between Windhoek and Johannesburg FIR's.

UT396 Bi-directional

POSITION	COMMENT	LATITUDE	LONGITUDE
UMTOL	FIR/FIR	272955.15S	0203940.57E
UDLON	*X UQ11	274350.57S	0204847.32E
UPV	VOR	282406.31S	0211537.48E

5 DESCRIPTION OF ATS ROUTES SUSPENDED WITHIN AORRA AIRSPACE

5.1 ATS routes contained within the boundaries of the AORRA suspended as indicated below:

Johannesburg Oceanic FIR

UQ11 UBVER (275522S 0141741E) to OKTEL (280753.81S 0150000E)

UA405 ETOBO (233900S 0100000W) to OKDOG (330500.00S 0150000E)

UL224 ITGIV (325600S 0100000W) to ITMEK (341200.00S 0150000E)

UL211F MUNES (402000S 0100000W) to ITLIK (351600.00S 0145957E)

5.2 Aircraft may track via a flight plan preferred track between these gates. Prior to entering or after exiting the AORRA at a particular gate, aircraft are to comply with the fixed route structure associated with that particular entry or exit point or as instructed by ATC, and are required to flight plan accordingly.

5.3 Prior to entering or after exiting the AORRA from/to Angolan Oceanic airspace, operators are to flight plan so as to cross the Angolan Oceanic FIR boundary at the significant point established on the FIR boundary and which is associated with the fixed route on which they intended operating while in that airspace.

6 RNAV / RNP ROUTE STRUCTURE WITHIN SOUTH AFRICAN AIRSPACE

6.1 The South-African Civil Aviation Authority (CAA) and Air Traffic and Navigation Services Company (ATNS) have concluded the trial period of the RNAV/RNP Routes.

6.2 The following RNAV/RNP routes are now permanently operational:

a. Johannesburg to Cape Town

b. Johannesburg to Durban

c. Durban to Cape Town

d. Johannesburg to East London

e. Johannesburg to George

f. Johannesburg to Port Elizabeth

g. Durban to Port Elizabeth and Cape Town

h. Cape Town across the Atlantic Ocean to destination (Oceanic)

i. Johannesburg across the Atlantic Ocean to destination (Oceanic)

j. Johannesburg to Zimbabwe and Botswana

k. Johannesburg to Mozambique and the Middle East

l. Johannesburg to Mozambique and the Far East

m. Johannesburg to Swaziland

n. Johannesburg to Richards Bay

6.3 Operators are to take note that non-compulsory reporting points (waypoints) have been included in the vicinity of crossing points to assist ATC in setting up separations. These reporting points (waypoints) must be available in the FMS in the event that ATC requires a flight to report at the reporting point (waypoint) for separation purposes.

6.4 The RNAV/RNP route structure will be applicable from FL200 and above.

6.4.1 Operators are to note that the RNAV are designed to RNP4 criteria, however aircraft which are capable of operating to RNP5 criteria will be permitted to operate on the routes provided there is continuous radar coverage.

6.4.2 Operators are advised that the next phase will be the implementation of RNP4 on the above mentioned routes. This will have an effect on the accuracy requirement of the airborne navigation equipment and operators who wish to continue to utilise these routes should take note of the following items.

a. REFERENCE MATERIAL:

The latest publication of ICAO document 9613 AN/937 - Manual on Required Navigation Performance (RNP).

b. REFERENCE MATERIAL:

RNP may be specified for a route, a number of routes, an area, volume of airspace or any airspace of defined dimensions (ICAO Doc 9613 AN/937-2.3.1).

c. AIRCRAFT NAVIGATION PERFORMANCE:

The concept of RNP is based on the expected navigation performance accuracy of the population of aircraft using the airspace. This in turn places demands on individual aircraft,

manufacturers of aircraft and aircraft operators to achieve the navigation performance required for a specific RNP type airspace on each flight. (ICAO DOC 9613-2.4.1).

d. CONTAINMENT PROBABILITY:

RNP types for en-route operations are established according to navigation performance accuracy in the horizontal plane, that is, lateral and longitudinal position fixing. The containment value is the distance from the intended position within which flights would be found for at least 95% of the total flight time.

e. REQUIRED NAVIGATION PERFORMANCE (RNP) TYPES.

RNP types (ICAO Doc 9613 AN/937-3.3.2) The RNP types most commonly used are 1.0, 4.0, 10, 12.6 and 20. The RNP type to be used within the South African domestic RNAV route structure will be RNP4. (Ref. Para 6.4.1 and 6.4.2).

f. AIRWORTHINESS APPROVAL FOR RNAV/RNP OPERATION

(See ICAO Doc 9613 AN/937 Chapter 5 & 6 and RSA AIP ENR 1.3) Operators who wish to make use of the benefits offered by RNAV operation will have to ensure that the navigation equipment required for RNAV/RNP use, installed in their aircraft, is approved by the Airworthiness section of the CAA. (TEL: (011) 545 1129, Contact Person: Carel Raath, or email at raathc@caa.co.za.) This includes onboard equipment associated with VOR/DME, which may be used, where the facilities are within range of the route flown.

g. BENEFITS OF RNAV/RNP ROUTES:

More direct routes permitting a reduction in flight distance.

Dual or parallel routes to accommodate a greater flow of enroute traffic.

Bypass routes for aircraft over flying high-density terminal areas.

Alternate or contingency routes on either a planned or an ad hoc basis.

Optimum locations for holding patterns.

Reduction in the number of ground navigation facilities.

h. FLIGHT CREW TRAINING REQUIREMENTS

(See ICAO Doc 9613 AN/937 Chapter 6-6.2/6.2.1-6.2.4). The training requirements of operators in respect of equipment and operating procedures should be adequately covered in the relevant operations and training manuals, where available. As a minimum, States should include training in equipment and operating procedures in pilots' training syllabi, such as instrument ratings, aircraft type ratings and refresher training.

6.5 The training should ensure that flight crews:

a. have a general knowledge of the application of RNP;

b. have a thorough understanding of the equipment;

c. are aware of equipment limitations;

d. have been trained in the operating procedures and safeguards necessary to obtain optimum efficiency and maintenance of required navigational accuracy;

e. are in current practise and have received recent training on the equipment;

f. appreciate the need to advise ATC should the accuracy of their navigation be in doubt; and

g. are conversant with contingency procedures.

Flight crew will have to prove competence in accordance with chapter 6 6.2.3 and satisfy the CAA.

6.6 AIR TRAFFIC CONTROL TRAINING

Air Traffic Controllers, will be required to be competent in accordance with ICAO Doc 9613 AN/937 Chapter 6 (6.2.5/6/7).

From the ATC point of view, the handling of traffic along RNP fixed and contingency routes will not be changed significantly.

The introduction of RNP areas including random tracks may bring about changes to the operation of ATC which would make it essential that additional training be provided, taking into account matters such as:

a. potentially different RNP type routes in the same sector;

b. transition between different RNP type areas;

c. radiotelephony (RTF) procedures (see 6.3);

d. revised military/civil and civil/civil co-ordination procedures;

e. conflict prediction and resolution along unpublished tracks; and

f. revised contingency procedures."

As more sophisticated navigation applications become more widely used (e.g. parallel offset capability, RNAV standard instrument departure (SID), and standard instrument arrivals (STAR), holding and approaches), their integration into ATC procedures will require that controllers are trained to accept and exploit the use of these advanced capabilities."

6.7 ATC NOTIFICATION

Pilots and/or operators intending to utilize the RNAV/RNP route structure are to flight plan appropriately.

7 . UPR GEOGRAPHICAL ZONE INSPIRE

7.1 Introduction

7.1.1 The Indian Ocean Strategic Partnership to Reduce Emissions (INSPIRE) is a partnership between Air Traffic and Navigation Services (ATNS) of South Africa, Air services Australia and Airports Authority of India. These partners are collaborating with peer Air Navigation Service Providers (ANSPs) across the Arabian Sea and Indian Ocean and a number of airline partners, to establish a User Preferred Route (UPR) Geographic Zone that will make UPRs available across the entire Indian Ocean area.

7.1.2 A User Preferred Route (UPR) during the oceanic phase of flight is defined as a lateral profile developed for each individual flight by the flight operator. These lateral profiles are customized in order to meet the specific needs of the aircraft operator for that flight, such as fuel optimisation, cost-index performance, or specific mission requirements.

7.1.3 Typically a UPR will be calculated by an aircraft operator's flight dispatch based on factors such as forecasted winds, aircraft type and performance, convective weather and scheduling requirements.

7.1.4 UPR's are a favoured enhancement to oceanic operations where air traffic control (ATC) limitations previously required that aircraft fly on fixed air traffic service (ATS) routes, or published flexible track systems. This enhancement is directly attributable to the implementation of ground and airborne improvements such as automated conflict prediction, conformance monitoring and automatic dependent surveillance (ADS).

7.1.5 When UPR's are created based on fuel optimisation considerations, the corresponding savings in greenhouse gas emissions can be substantial.

7.1.6 ATNS, with INSPIRE partners and peer ANSPs, are participating in a UPR trial program as part of INSPIRE.

7.1.7 IORRA and associated procedures are cancelled and replaced by UPR Geographical Zone.

7.2 Procedures

7.2.1 INSPIRE airline partners that have received prior approval from INSPIRE, may use the following procedures within the Johannesburg Oceanic Flight Information Region (FAJO) Indian Ocean:

7.2.2 Flights must enter or exit the FAJO FIR via published waypoints or position of whole degrees of latitude and longitude described in degrees and minutes.

7.2.3 UP's must be constructed via published waypoints, navigation aids, or whole degrees of latitude and longitude.

7.2.4 UPRs may include air routes.

7.3 Access to UPR

7.3.1 Airspace users may only flight plan a User Preferred Route (UPR) Geographic Zone if they meet the following minimum criteria:

a. RNAV10, and

b. ADS-C/CPDLC equipped.

Note: The minimum criteria listed above must be notified in the Flight Plan.

7.3.2 Suspension of UPR Operations. In the event of a contingency situation, the UPR Operations shall be suspended.

7.4 Contacts

Any questions and assistance should be directed to:

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RNAV ROUTES AND WAYPOINTS

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Q56	ENR 3.3-83	UQ56	ENR 3.3-84						
Q58	ENR 3.3-84	UQ58	ENR 3.3-85						
Q59	ENR 3.3-93	UQ59	ENR 3.3-93						
Q60	ENR 3.3-93	UQ60	ENR 3.3-93						
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Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Q10						
▲RAGUL 262654.35S 0273443.89E						
	242° 42 NM	FL245 FL200 Class A	10		↓	
▲UTEBA 265711.00S 0270214.28E						
	244° 89 NM	FL245 FL200 Class A	10		↓	
▲APMIN 280038.45S 0255128.05E						
	245° 50 NM	FL245 FL200 Class A	10		↓	
▲EVESI 283554.70S 0251112.70E						
	246° 199 NM	FL245 FL200 Class A	10		↓	
▲EGNOM 305403.37S 0222635.25E						
	250° 188 NM	FL245 FL200 Class A	10		↓	
▲NOKOX 330114.23S 0194303.34E						
	250° 20 NM	FL245 FL200 Class A	10		↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

UQ10						
▲RAGUL 262654.35S 0273443.89E						
	242° 42 NM	FL460 FL245 Class A	10		↓	
▲UTEBA 265711.00S 0270214.28E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	244° 89 NM	FL460 FL245 Class A	10		↓	
▲APMIN 280038.45S 0255128.05E						
	245° 50 NM	FL460 FL245 Class A	10		↓	
▲EVESI 283554.70S 0251112.70E						
	246° 199 NM	FL460 FL245 Class A	10		↓	
▲EGNOM 305403.37S 0222635.25E						
	250° 188 NM	FL460 FL245 Class A	10		↓	
▲NOKOX 330114.23S 0194303.34E						
	250° 20 NM	FL460 FL245 Class A	10		↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

Z2						
▲TETAN 333948.00S 0192458.00E						
	077° 36 NM	FL245 FL200 Class A	10	↓		
▲AXOPA 331742.26S 0195833.41E						
	073° 200 NM	FL245 FL200 Class A	10	↓		
▲EGTIL 310223.29S 0225155.48E						
	70° 230 NM	FL245 FL200 Class A	10	↓		
▲UNRAG 282149.12S 0260109.16E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	066° 30 NM	FL245 FL200 Class A	10	↓		
▲APRAX 280031.67S 0262508.12E						
	065° 68 NM	FL245 FL200 Class A	10	↓		
▲TEVAR 271221.37S 0271833.24E						
	060° 30 NM	FL245 FL200 Class A	10	↓		
▲NIBEX 264925.56S 0274013.41E						

UZ2						
▲TETAN 333948.00S 0192458.00E						
	077° 36 NM	FL 460 FL 245 Class A	10	↓		
▲AXOPA 331742.26S 0195833.41E						
	073° 200 NM	FL 460 FL 245 Class A	10	↓		
▲EGTIL 310223.29S 0225155.48E						
	70° 230 NM	FL 460 FL 245 Class A	10	↓		
▲UNRAG 282149.12S 0260109.16E						
	066° 30 NM	FL 460 FL 245 Class A	10	↓		
▲APRAX 280031.67S 0262508.12E						
	065° 68 NM	FL 460 FL 245 Class A	10	↓		
▲TEVAR 271221.37S 0271833.24E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	060º 30 NM	FL 460 FL 245 Class A	10	↓		
▲NIBEX 264925.56S 0274013.41E						

Q7						
▲APDAK 263303.54S 0290252.64E						
	212º 51 NM	FL245 FL200 Class A	10		↓	
▲GEROX 272249.01S 0284943.11E						
	195º 36 NM	FL245 FL200 Class A	10		↓	
▲IXARA 275827.43S 0285340.08E						
	212º 154 NM	FL245 FL200 Class A	10		↓	
▲OKBUS 302947.44S 0281845.93E						
	216º 25 NM	FL 245 FL 200 Class A	10		↓	
▲ETLUR 305425.43S 0281254.69E						
	217º 24 NM	FL245 FL200 Class A	10		↓	
▲APMUL 311747.38S 0280718.59E						
	217º 13 NM	FL245 FL200 Class A	10		↓	
▲TIDAX 313034.55S 0280413.34E						
	217º 42 NM	FL245 FL200 Class A	10		↓	
▲AXOXI 321215.60S 0275404.32E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ7						
▲APDAK 263303.54S 0290252.64E						
	212º 51 NM	FL460 FL245 Class A	10		↓	
▲GEROX 272249.01S 0284943.11E						
	195º 36 NM	FL460 FL245 Class A	10		↓	
▲IXARA 275827.43S 0285340.08E						
	212º 154 NM	FL460 FL245 Class A	10		↓	
▲OKBUS 302947.44S 0281845.93E						
	216º 25 NM	FL460 FL245 Class A	10		↓	
▲ETLUR 305425.43S 0281254.69E						
	217º 24 NM	FL460 FL245 Class A	10		↓	
▲APMUL 311747.38S 0280718.59E						
	217º 13 NM	FL460 FL245 Class A	10		↓	
▲TIDAX 313034.55S 0280413.34E						
	217º 42 NM	FL460 FL245 Class A	10		↓	
▲AXOXI 321215.60S 0275404.32E						

Q9						
▲NEVEN 272859.15S 0280257.10E						
	236º 38 NM	FL245 FL200 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲XOTOP 275948.66S 0273806.53E						
	235º 59 NM	FL245 FL200 Class A	10		↓	
▲NIDIG 284806.79S 0270018.48E						
	236º 168 NM	FL245 FL200 Class A	10		↓	
▲ETGIS 310511.89S 0250856.95E						
	240º 100 NM	FL245 FL200 Class A	10		↓	
▲ESVAX 322602.43S 0240006.71E						
	241º 38 NM	FL245 FL200 Class A	10		↓	
▲GERUT 325639.54S 0233320.68E						
	242º 58 NM	FL245 FL200 Class A	10		↓	
▲XALIN 334308.69S 0225201.48E						

UQ9						
▲NEVEN 272859.15S 0280257.10E						
	236º 38 NM	FL460 FL245 Class A	10		↓	
▲XOTOP 275948.66S 0273806.53E						
	235º 59 NM	FL460 FL245 Class A	10		↓	
▲NIDIG 284806.79S 0270018.48E						
	236º 168 NM	FL460 FL245 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲ETGIS 310511.89S 0250856.95E						
	240º 100 NM	FL460 FL245 Class A	10		↓	
▲ESVAX 322602.43S 0240006.71E						
	241º 38 NM	FL460 FL245 Class A	10		↓	
▲GERUT 325639.54S 0233320.68E						
	242º 58 NM	FL460 FL245 Class A	10		↓	
▲XALIN 334308.69S 0225201.48E						

Z4						
▲KATOR 332919.46S 0223834.37E						
	062º 25 NM	FL245 FL200 Class A	10	↓		
▲XANDA 330859.08S 0225623.03E						
	062º 38 NM	FL245 FL200 Class A	10	↓		
▲AVEXA 323836.27S 0232245.08E						
	061º 108 NM	FL245 FL200 Class A	10	↓		
▲NIDOX 311058.78S 0243646.21E						
	059º 198 NM	FL245 FL200 Class A	10	↓		
▲OKTIG 282838.68S 0264640.84E						
	055º 34 NM	FL245 FL200 Class A	10	↓		
▲EGSAN 280010.72S 0270835.76E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	031° 49 NM	FL245 FL200 Class A	10	↓		
▲TEVAR 271221.37S 0271833.24E						

UZ4						
▲KATOR 332919.46S 0223834.37E						
	062° 25 NM	FL460 FL245 Class A	10	↓		
▲XANDA 330859.08S 0225623.03E						
	062° 38 NM	FL460 FL245 Class A	10	↓		
▲AVEXA 323836.27S 0232245.08E						
	061° 108 NM	FL460 FL245 Class A	10	↓		
▲NIDOX 311058.78S 0243646.21E						
	059° 198 NM	FL460 FL245 Class A	10	↓		
▲OKTIG 282838.68S 0264640.84E						
	055° 34 NM	FL460 FL245 Class A	10	↓		
▲EGSAN 280010.72S 0270835.76E						
	031° 49 NM	FL460 FL245 Class A	10	↓		
▲TEVAR 271221.37S 0271833.24E						

Q12		
▲OKTED 341312.00S 0193330.00E		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	111° 32 NM	FL245 FL200 Class A	10	↓		
▲XUGOL 341034.13S 0201127.59E						
	091° 27 NM	FL245 FL200 Class A	10	↓		
▲APKAP 335918.93S 0204131.63E						
	091° 16 NM	FL245 FL200 Class A	10	↓		
▲UNRES 335243.30S 0205856.24E						
	091° 79 NM	FL245 FL200 Class A	10	↓		
▲EXELO 331930.39S 0222426.38E						
	094° 29 NM	FL245 FL200 Class A	10	↓		
▲XANDA 330859.08S 0225623.03E						
	094° 33 NM	FL245 FL200 Class A	10	↓		
▲GERUT 325639.54S 0233320.68E						
	094° 153 NM	FL245 FL200 Class A	10	↓		
▲IXEBI 315753.02S 0262047.33E						
	092° 26 NM	FL245 FL200 Class A	10	↓		
▲OKPUN 314733.63S 0264840.64E						
	092° 59 NM	FL245 FL200 Class A	10	↓		
▲DUDVU 312352.27S 0275136.50E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	091° 15 NM	FL245 FL200 Class A	10	↓		
▲APMUL 311747.38S 0280718.59E						
	091° 42 NM	FL245 FL200 Class A	10	↓		
▲OKLOL 310028.20S 0285145.51E						
	091° 66 NM	FL245 FL200 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						
	078° 30 NM	FL245 FL200 Class A	10	↓		
▲DUNSA 301406.71S 0302914.71E						

UQ12						
▲OKTED 341312.00S 0193330.00E						
	111° 32 NM	FL460 FL245 Class A	10	↓		
▲XUGOL 341034.13S 0201127.59E						
	091° 27 NM	FL460 FL245 Class A	10	↓		
▲APKAP 335918.93S 0204131.63E						
	091° 16 NM	FL460 FL245 Class A	10	↓		
▲UNRES 335243.30S 0205856.24E						
	091° 79 NM	FL460 FL245 Class A	10	↓		
▲EXELO 331930.39S 0222426.38E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	094° 29 NM	FL460 FL245 Class A	10	↓		
▲XANDA 330859.08S 0225623.03E						
	094° 33 NM	FL460 FL245 Class A	10	↓		
▲GERUT 325639.54S 0233320.68E						
	094° 153 NM	FL460 FL245 Class A	10	↓		
▲IXEBI 315753.02S 0262047.33E						
	092° 26 NM	FL460 FL245 Class A	10	↓		
▲OKPUN 314733.63S 0264840.64E						
	092° 59 NM	FL460 FL245 Class A	10	↓		
▲DUDVU 312352.27S 0275136.50E						
	091° 15 NM	FL460 FL245 Class A	10	↓		
▲APMUL 311747.38S 0280718.59E						
	091° 42 NM	FL460 FL245 Class A	10	↓		
▲OKLOL 310028.20S 0285145.51E						
	091° 66 NM	FL460 FL245 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						
	078° 30 NM	FL460 FL245 Class A	10	↓		
▲DUNSA 301406.71S 0302914.71E						

Route designator Name of significant points Coordinates	*Route MAG Track MAG DIST*	*Upper limit Lower limit Classification*	*Lateral limits NM*	*Direction of cruising levels*		*Remarks Controlling unit Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Q13						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	152º 48 NM	FL245 FL200 Class A	10	↓		
▲UVITI 255542.74S 0314511.76E						

UQ13						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	152º 48 NM	FL460 FL245 Class A	10	↓		
▲UVITI 255542.74S 0314511.76E						

Z3						
▲TUBIN 300201.29S 0301754.66E						
	265º 30 NM	FL245 FL200 Class A	10		↓	
▲NAMGU 301704.14S 0294758.45E						
	270º 69 NM	FL245 FL200 Class A	10		↓	
▲EVUVA 304535.88S 0283542.17E						
	271º 22 NM	FL245 FL200 Class A	10		↓	
▲ETLUR 305425.43S 0281254.69E						
	271º 16 NM	FL245 FL200 Class A	10		↓	
▲XADUP 310048.50S 0275613.14E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	271° 54 NM	FL245 FL200 Class A	10		↓	
▲IXAKU 312230.06S 0265846.63E						
	272° 26 NM	FL245 FL200 Class A	10		↓	
▲XABDO 313246.19S 0263047.81E						
	272° 139 NM	FL245 FL200 Class A	10		↓	
▲ESVAX 322602.43S 0240006.71E						
	273° 34 NM	FL245 FL200 Class A	10		↓	
▲AVEXA 323836.27S 0232245.08E						
	274° 45 NM	FL245 FL200 Class A	10		↓	
▲ETKAL 325448.40S 0223336.03E						
	270° 81 NM	FL245 FL200 Class A	10		↓	
▲GADNU 332851.57S 0210631.19E						
	271° 71 NM	FL245 FL200 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

UZ3						
▲TUBIN 300201.29S 0301754.66E						
	265° 30 NM	FL460 FL245 Class A	10		↓	
▲NAMGU 301704.14S 0294758.45E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	270º 69 NM	FL460 FL245 Class A	10		↓	
▲EVUVA 304535.88S 0283542.17E						
	271º 22 NM	FL460 FL245 Class A	10		↓	
▲ETLUR 305425.43S 0281254.69E						
	271º 16 NM	FL460 FL245 Class A	10		↓	
▲XADUP 310048.50S 0275613.14E						
	271º 54 NM	FL460 FL245 Class A	10		↓	
▲IXAKU 312230.06S 0265846.63E						
	272º 26 NM	FL460 FL245 Class A	10		↓	
▲XABDO 313246.19S 0263047.81E						
	272º 139 NM	FL460 FL245 Class A	10		↓	
▲ESVAX 322602.43S 0240006.71E						
	273º 34 NM	FL460 FL245 Class A	10		↓	
▲AVEXA 323836.27S 0232245.08E						
	274º 45 NM	FL460 FL245 Class A	10		↓	
▲ETKAL 325448.40S 0223336.03E						
	270º 81 NM	FL460 FL245 Class A	10		↓	
▲GADNU 332851.57S 0210631.19E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	271° 71 NM	FL460 FL245 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

Q8						
▲NEVEN 272859.15S 0280257.10E						
	218° 32 NM	FL245 FL200 Class A	10		↓	
▲ATUXO 275935.97S 0275207.27E						
	219° 33 NM	FL245 FL200 Class A	10		↓	
▲IXEKI 283043.45S 0274036.85E						
	220° 142 NM	FL245 FL200 Class A	10		↓	
▲EGTAP 304616.63S 0264903.41E						
	223° 49 NM	FL245 FL200 Class A	10		↓	
▲XABDO 313246.19S 0263047.81E						
	224° 26 NM	FL245 FL200 Class A	10		↓	
▲IXEBI 315753.02S 0262047.33E						
	224° 28 NM	FL245 FL200 Class A	10		↓	
▲TIKER 322402.12S 0261016.18E						
	225° 28 NM	FL245 FL200 Class A	10		↓	
▲XOXOR 325008.37S 0255939.92E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	225° 22 NM	FL245 FL200 Class A	10		↓	
▲ALKAT 331100.54S 0255107.43E						

UQ8						
▲NEVEN 272859.15S 0280257.10E						
	218° 32 NM	FL460 FL245 Class A	10		↓	
▲ATUXO 275935.97S 0275207.27E						
	219° 33 NM	FL460 FL245 Class A	10		↓	
▲IXEKI 283043.45S 0274036.85E						
	220° 142 NM	FL460 FL245 Class A	10		↓	
▲EGTAP 304616.63S 0264903.41E						
	223° 49 NM	FL460 FL245 Class A	10		↓	
▲XABDO 313246.19S 0263047.81E						
	224° 26 NM	FL460 FL245 Class A	10		↓	
▲IXEBI 315753.02S 0262047.33E						
	224° 28 NM	FL460 FL245 Class A	10		↓	
▲TIKER 322402.12S 0261016.18E						
	225° 28 NM	FL460 FL245 Class A	10		↓	
▲XOXOR 325008.37S 0255939.92E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	225° 22 NM	FL460 FL245 Class A	10		↓	
▲ALKAT 331100.54S 0255107.43E						

Z5						
▲ITKIT 332137.27S 0260950.41E						
	047° 43 NM	FL245 FL200 Class A	10	↓		
▲XARVI 324125.16S 0262636.17E						
	046° 32 NM	FL245 FL200 Class A	10	↓		
▲IMROG 321042.91S 0263914.43E						
	045° 24 NM	FL245 FL200 Class A	10	↓		
▲OKPUN 314733.63S 0264840.64E						
	045° 26 NM	FL245 FL200 Class A	10	↓		
▲IXAKU 312230.06S 0265846.63E						
	044° 44 NM	FL245 FL200 Class A	10	↓		
▲EVEMU 304104.75S 0271517.92E						
	043° 151 NM	FL245 FL200 Class A	10	↓		
▲XORER 281735.22S 0281037.24E						
	040° 19 NM	FL245 FL200 Class A	10	↓		
▲IMGEV 275909.30S 0281732.10E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	038º 32 NM	FL245 FL200 Class A	10	↓		
▲OKTEK 272835.55S 0282748.16E						
	045º 33 NM	FL245 FL200 Class A	10	↓		
▲AVILO 265854.69S 0284315.62E						
	044º 19 NM	FL245 FL200 Class A	10	↓		
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						

UZ5						
▲ITKIT 332137.27S 0260950.41E						
	047º 43 NM	FL460 FL245 Class A	10	↓		
▲XARVI 324125.16S 0262636.17E						
	046º 32 NM	FL460 FL245 Class A	10	↓		
▲IMROG 321042.91S 0263914.43E						
	045º 24 NM	FL460 FL245 Class A	10	↓		
▲OKPUN 314733.63S 0264840.64E						
	045º 26 NM	FL460 FL245 Class A	10	↓		
▲IXAKU 312230.06S 0265846.63E						
	044º 44 NM	FL460 FL245 Class A	10	↓		
▲EVEMU 304104.75S 0271517.92E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	043º 151 NM	FL460 FL245 Class A	10	↓		
▲XORER 281735.22S 0281037.24E						
	040º 19 NM	FL460 FL245 Class A	10	↓		
▲IMGEV 275909.30S 0281732.10E						
	038º 32 NM	FL460 FL245 Class A	10	↓		
▲OKTEK 272835.55S 0282748.16E						
	045º 33 NM	FL460 FL245 Class A	10	↓		
▲AVILO 265854.69S 0284315.62E						
	044º 19 NM	FL460 FL245 Class A	10	↓		
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						

Q14						
▲UVIKU 332734.47S 0173425.87E						
	355° 173° 201 NM	FL245 FL200 Class A	10	↑	↓	
▲IMLUT 303035.90S 0154102.74E						
	357° 176° 88 NM	FL245 FL200 Class A	10	↑	↓	
▲UVGOD 290943.27S 0150000.00E						

UQ14		
▲UVIKU 332734.47S 0173425.87E		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	355° 173° 201 NM	FL460 FL245 Class A	10	↑	↓	
▲IMLUT 303035.90S 0154102.74E						
	357° 176° 88 NM	FL460 FL245 Class A	10	↑	↓	
▲UVGOD 290943.27S 0150000.00E						

Q15						
▲APKIN 340500.64S 0170059.32E						
	324° 144° 117 NM	FL245 FL200 Class A	10	↑	↓	
▲OKDOG 330500.00S 0150000.00E						

UQ15						
▲APKIN 340500.64S 0170059.32E						
	324° 144° 117 NM	FL460 FL245 Class A	10	↑	↓	
▲OKDOG 330500.00S 0150000.00E						

Q19						
▲AVUSA 272955.10S 0221330.09E						
	086º 192 NM	FL245 FL200 Class A	10	↓		
▲TEXIP 261303.77S 0253025.47E						
	083º 45 NM	FL245 FL200 Class A	10	↓		
▲ETBUX 255423.80S 0261558.08E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels* *Odd*	*Even*	*Remarks* *Controlling unit* *Frequency*
1	***2***	***3***	***4***	***5***		***6***
	083° 35 NM	FL245 FL200 Class A	10	↓		
▲PEDIL 253933.67S 0265134.05E						
	114° 32 NM	FL245 FL200 Class A	10	↓		
▲AVAGO 254309.90S 0272639.36E						

UQ19						
▲AVUSA 272955.10S 0221330.09E						
	086° 192 NM	FL460 FL245 Class A	10	↓		
▲TEXIP 261303.77S 0253025.47E						
	083° 45 NM	FL460 FL245 Class A	10	↓		
▲ETBUX 255423.80S 0261558.08E						
	083° 35 NM	FL460 FL245 Class A	10	↓		
▲PEDIL 253933.67S 0265134.05E						
	114° 32 NM	FL460 FL245 Class A	10	↓		
▲AVAGO 254309.90S 0272639.36E						

Q11						
▲AVUSA 272955.10S 0221330.09E						
	278° 098° 76.5 NM	FL245 FL200 Class A	10	↑	↓	
▲UDLON 274350.57S 0204847.32E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	278° 099° 118.7 NM	FL245 FL200 Class A	10	↑	↓	
▲ITKOT 280341.17S 0183640.38E						
	273° 093° 14.3 NM	FL245 FL200 Class A	10	↑	↓	
▲EGTAR 280738.17S 0182105.08E						
	273° 093° 12.8 NM	FL245 FL200 Class A	10	↑	↓	
▲ETUPO 281107.96S 0180709.80E						
	273° 093° 12.7 NM	FL245 FL200 Class A	10	↑	↓	
▲ITNOX 281434.29S 0175321.38E						
	273° 093° 14.9 NM	FL245 FL200 Class A	10	↑	↓	
▲EKBON 281834.94S 0173706.30E						
	273° 094° 44.6 NM	FL245 FL200 Class A	10	↑	↓	
▲ANTIL 283022.76S 0164820.05E						
	274° 094° 14.9 NM	FL245 FL200 Class A	10	↑	↓	
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	307° 127° 14.9 NM	FL245 FL200 Class A	10	↑	↓	
▲PAREK 282943.84S 0161557.51E						
	307° 127° 70.5 NM	FL245 FL200 Class A	10	↑	↓	
▲OKTEL 280753.81S 0150000.00E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	307° 127° 24.5 NM	FL245 FL200 Class A	10	↑	↓	
▲ANVAR 280009.00S 0143344.45E						
	307° 127° 15.0 NM	FL245 FL200 Class A	10	↑	↓	
▲UBVER 275521.71S 0141740.93E						
	307° 127° 15.0 NM	FL245 FL200 Class A	10	↑	↓	
▲EGRAP 275037.51S 0140136.97E						

UQ11						
▲AVUSA 272955.10S 0221330.09E						
	278° 098° 76.5 NM	FL460 FL245 Class A	10	↑	↓	
▲UDLON 274350.57S 0204847.32E						
	278° 099° 118.7 NM	FL460 FL245 Class A	10	↑	↓	
▲ITKOT 280341.17S 0183640.38E						
	273° 093° 14.3 NM	FL460 FL245 Class A	10	↑	↓	
▲EGTAR 280738.17S 0182105.08E						
	273° 093° 12.8 NM	FL460 FL245 Class A	10	↑	↓	
▲ETUPO 281107.96S 0180709.80E						
	273° 093° 12.7 NM	FL460 FL245 Class A	10	↑	↓	
▲ITNOX 281434.29S 0175321.38E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	273º 093º 14.9 NM	FL460 FL245 Class A	10	↑	↓	
▲EKBON 281834.94S 0173706.30E						
	273º 094º 44.6 NM	FL460 FL245 Class A	10	↑	↓	
▲ANTIL 283022.76S 0164820.05E						
	274º 094º 14.9 NM	FL460 FL245 Class A	10	↑	↓	
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	307º 127º 14.9 NM	FL460 FL245 Class A	10	↑	↓	
▲PAREK 282943.84S 0161557.51E						
	307º 127º 70.5 NM	FL460 FL245 Class A	10	↑	↓	
▲OKTEL 280753.81S 0150000.00E						
	307º 127º 24.5 NM	FL460 FL245 Class A	10	↑	↓	
▲ANVAR 280009.00S 0143344.45E						
	307º 127º 15.0 NM	FL460 FL245 Class A	10	↑	↓	
▲UBVER 275521.71S 0141740.93E						
	307º 127º 15.0 NM	FL460 FL245 Class A	10	↑	↓	
▲EGRAP 275037.51S 0140136.97E						

Q1		
▲NESAN 252948.88S 0280705.56E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	027º 41 NM	FL245 FL200 Class A	10	↓		
▲IMKUP 244915.70S 0281420.65E						
	018º 72 NM	FL245 FL200 Class A	10	↓		
▲IMPOD 233643.21S 0281607.85E						
	016º 59 NM	FL245 FL200 Class A	10	↓		
▲TAVLA 223724.00S 0281736.00E						

UQ1						
▲NESAN 252948.88S 0280705.56E						
	027º 41 NM	FL460 FL245 Class A	10	↓		
▲IMKUP 244915.70S 0281420.65E						
	018º 72 NM	FL460 FL245 Class A	10	↓		
▲IMPOD 233643.21S 0281607.85E						
	016º 59 NM	FL460 FL245 Class A	10	↓		
▲TAVLA 223724.00S 0281736.00E						

Z6						
▲RAGUL 262654.35S 0273443.89E						
	274º 55 NM	FL245 FL200 Class A	10		↓	
▲EXOKU 264010.40S 0263440.15E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	***2***	***3***	***4***	***5***		***6***
	275º 62 NM	FL245 FL200 Class A	10		↓	
▲UNGEK 265428.31S 0252726.37E						
	276º 176 NM	FL245 FL200 Class A	10		↓	
▲AVUSA 272955.10S 0221330.09E						

UZ6						
▲RAGUL 262654.35S 0273443.89E						
	274º 55 NM	FL460 FL245 Class A	10		↓	
▲EXOKU 264010.40S 0263440.15E						
	275º 62 NM	FL460 FL245 Class A	10		↓	
▲UNGEK 265428.31S 0252726.37E						
	276º 176 NM	FL460 FL245 Class A	10		↓	
▲AVUSA 272955.10S 0221330.09E						

Z7						
▲TAVLA 223724.00S 0281736.00E						
	179º 74 NM	FL245 FL200 Class A	10		↓	
▲UTIGU 234947.99S 0283708.32E						
	184º 68 NM	FL245 FL200 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	208° 31 NM	FL245 FL200 Class A	10		↓	
▲OKPIT 252744.22S 0284432.56E						

UZ7						
▲TAVLA 223724.00S 0281736.00E						
	179° 74 NM	FL460 FL245 Class A	10		↓	
▲UTIGU 234947.99S 0283708.32E						
	184° 68 NM	FL460 FL245 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						
	208° 31 NM	FL460 FL245 Class A	10		↓	
▲OKPIT 252744.22S 0284432.56E						

Z8						
▲AVAVA 265456.32S 0292710.11E						
	147° 63 NM	FL245 FL200 Class A	10	↓		
▲ETMAL 273422.09S 0302300.51E						
	148° 114 NM	FL245 FL200 Class A	10	↓		
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						

UZ8		
▲AVAVA 265456.32S 0292710.11E		

Route designator Name of significant points Coordinates	*Route MAG Track MAG DIST*	*Upper limit Lower limit Classification*	*Lateral limits NM*	*Direction of cruising levels*		*Remarks Controlling unit Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	147° 63 NM	FL460 FL245 Class A	10	↓		
▲ETMAL 273422.09S 0302300.51E						
	148° 114 NM	FL460 FL245 Class A	10	↓		
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						

Z9						
▲ALENI 265835.54S 0320937.89E						
	301° 75 NM	FL245 FL200 Class A	10		↓	
▲APNON 264537.86S 0304652.79E						
	299° 62 NM	FL245 FL200 Class A	10		↓	
▲TETIK 263414.18S 0293825.33E						
	278° 42 NM	FL245 FL200 Class A	10		↓	
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						

UZ9						
▲ALENI 265835.54S 0320937.89E						
	301° 75 NM	FL460 FL245 Class A	10		↓	
▲APNON 264537.86S 0304652.79E						
	299° 62 NM	FL460 FL245 Class A	10		↓	
▲TETIK 263414.18S 0293825.33E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	278° 42 NM	FL460 FL245 Class A	10		↓	
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						
Z10						
▲ETOSA 244250.40S 0260312.98E						
	158° 37 NM	FL245 FL200 Class A	10	↓		
▲UDBUL 251203.81S 0262800.52E						
	158° 35 NM	FL245 FL200 Class A	10	↓		
▲PEDIL 253933.67S 0265134.05E						
UZ10						
▲ETOSA 244250.40S 0260312.98E						
	158° 37 NM	FL460 FL245 Class A	10	↓		
▲UDBUL 251203.81S 0262800.52E						
	158° 35 NM	FL460 FL245 Class A	10	↓		
▲PEDIL 253933.67S 0265134.05E						
Z11						
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						
	204° 112 NM	FL245 FL200 Class A	10		↓	
▲DUMGO 240510.31S 0290150.64E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	206° 52 NM	FL245 FL200 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						

UZ11						
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						
	204° 112 NM	FL460 FL245 Class A	10		↓	
▲DUMGO 240510.31S 0290150.64E						
	206° 52 NM	FL460 FL245 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						

Z13						
▲XUGOL 341034.13S 0201127.59E						
	111° 109 NM	FL245 FL200 Class A	10	↓		
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	109° 224 NM	FL245 FL200 Class A	10	↓		
▲XAKIM 332622.65S 0264747.86E						
	108° 7 NM	FL245 FL200 Class A	10	↓		
▲ETKON 332508.00S 0265621.52E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UZ13						
▲XUGOL 341034.13S 0201127.59E						
	111° 109 NM	FL460 FL245 Class A	10	↓		
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	109° 224 NM	FL460 FL245 Class A	10	↓		
▲XAKIM 332622.65S 0264747.86E						
	108° 7 NM	FL460 FL245 Class A	10	↓		
▲ETKON 332508.00S 0265621.52E						

Z16						
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	213° 49 NM	FL245 FL200 Class A	10		↓	
▲NEVEN 272859.15S 0280257.10E						

UZ16						
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	213° 49 NM	FL460 FL245 Class A	10		↓	
▲NEVEN 272859.15S 0280257.10E						

Z17						
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	163º 50 NM	FL245 FL200 Class A	10	↓		
▲GEROX 272249.01S 0284943.11E						

UZ17						
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	163º 50 NM	FL460 FL245 Class A	10	↓		
▲GEROX 272249.01S 0284943.11E						

Z18						
▲OVALA 264903.39S 0283744.70E						
	237º 50 NM	FL245 FL200 Class A	10		↓	
▲NEVEN 272859.15S 0280257.10E						

UZ18						
▲OVALA 264903.39S 0283744.70E						
	237º 50 NM	FL460 FL245 Class A	10		↓	
▲NEVEN 272859.15S 0280257.10E						

Z21						
▲VASUR 253346.76S 0275335.87E						
	338º 40 NM	FL245 FL200 Class A	10		↓	
▲ITROL 250207.97S 0272605.37E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UZ21						
▲VASUR 253346.76S 0275335.87E						
	338º 40 NM	FL460 FL245 Class A	10		↓	
▲ITROL 250207.97S 0272605.37E						

Z22						
▲EVIPI 245200.20S 0275054.35E						
	219º 56 NM	FL245 FL200 Class A	10		↓	
▲AVAGO 254309.90S 0272639.36E						

UZ22						
▲EVIPI 245200.20S 0275054.35E						
	219º 56 NM	FL460 FL245 Class A	10		↓	
▲AVAGO 254309.90S 0272639.36E						

Z23						
▲VASUR 253346.76S 0275335.87E						
	041º 48 NM	FL245 FL200 Class A	10	↓		
▲IMKUP 244915.70S 0281420.65E						

UZ23						
▲VASUR 253346.76S 0275335.87E						
	041º 48 NM	FL460 FL245 Class A	10	↓		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲IMKUP 244915.70S 0281420.65E						

Z26						
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						
	257º 16 NM	FL245 FL200 Class A	10		↓	
▲ERDAS 332450.10S 0191112.85E						

UZ26						
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						
	257º 16 NM	FL460 FL245 Class A	10		↓	
▲ERDAS 332450.10S 0191112.85E						

UQ2						
▲EGMEN 253845.74S 0285805.53E						
	065º 18 NM	FL460 FL245 Class A	10	↓		
▲UNDIT 252647.15S 0291216.04E						
	065º 53 NM	FL460 FL245 Class A	10	↓		
▲NEVUK 245020.43S 0295524.24E						
	065º 10 NM	FL460 FL245 Class A	10	↓		
▲XARMA 244331.61S 0300328.12E						
	064º 10 NM	FL460 FL245 Class A	10	↓		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲XATNU 243642.99S 0301131.77E						
	064° 102 NM	FL460 FL245 Class A	10	↓		
▲EPSEK 232715.00S 0313345.00E						

Q2						
▲EGMEN 253845.74S 0285805.53E						
	065° 18 NM	FL245 FL200 Class A	10	↓		
▲UNDIT 252647.15S 0291216.04E						
	065° 53 NM	FL245 FL200 Class A	10	↓		
▲NEVUK 245020.43S 0295524.24E						
	065° 10 NM	FL245 FL200 Class A	10	↓		
▲XARMA 244331.61S 0300328.12E						
	064° 10 NM	FL245 FL200 Class A	10	↓		
▲XATNU 243642.99S 0301131.77E						
	064° 102 NM	FL245 FL200 Class A	10	↓		
▲EPSEK 232715.00S 0313345.00E						

T122						
▲EXOBI 260252.27S 0290851.32E						
	106° 79 NM	FL245 FL200 Class A	10	↓		
▲ESVEV 255913.02S 0303619.18E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	106º 28 NM	FL245 FL200 Class A	10	↓		
▲IMKID 255743.38S 0310701.21E						
	106º 6 NM	FL245 FL200 Class A	10	↓		
▲DULPU 255723.71S 0311325.26E						
	106º 29 NM	FL245 FL200 Class A	10	↓		
▲UVITI 255542.74S 0314511.76E						
	106º 12 NM	FL245 FL200 Class A	10	↓		
▲ORNAD 255500.00S 0315800.00E						

UT122						
▲EXOBI 260252.27S 0290851.32E						
	106º 79 NM	FL460 FL245 Class A	10	↓		
▲ESVEV 255913.02S 0303619.18E						
	106º 28 NM	FL460 FL245 Class A	10	↓		
▲IMKID 255743.38S 0310701.21E						
	106º 6 NM	FL460 FL245 Class A	10	↓		
▲DULPU 255723.71S 0311325.26E						
	106º 29 NM	FL460 FL245 Class A	10	↓		
▲UVITI 255542.74S 0314511.76E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	106º 12 NM	FL460 FL245 Class A	10	↓		
▲ORNAD 255500.00S 0315800.00E						

T125						
▲ANVAK 254020.71S 0315950.23E						
	309º 21NM	FL245 FL200 Class A	10		↓	
▲GAGNO 253309.23S 0313749.16E						
	309º 30 NM	FL245 FL200 Class A			↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						

UT125						
▲ANVAK 254020.71S 0315950.23E						
	309º 21NM	FL460 FL245 Class A	10		↓	
▲GAGNO 253309.23S 0313749.16E						
	309º 30 NM	FL460 FL245 Class A			↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						

T433						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	322º 142º 70 NM	FL245 FL200 Class A	10	↑	↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲XARMA 244331.61S 0300328.12E						
	321º 141º 25 NM	FL245 FL200 Class A	10	↑	↓	
▲UTUPU 242925.25S 0294048.28E						
	320º 140º 43 NM	FL245 FL200 Class A	10	↑	↓	
▲DUMGO 240510.31S 0290150.64E						
	320º 139º 27 NM	FL245 FL200 Class A	10	↑	↓	
▲UTIGU 234947.99S 0283708.32E						
	319º 139º 23 NM	FL245 FL200 Class A	10	↑	↓	
▲IMPOD 233643.21S 0281607.85E						
	319º 138º 10 NM	FL245 FL200 Class A	10	↑	↓	
▲EKBAP 233100.95S 0280657.94E						
	318º 138º 32.46 NM	FL245 FL200 Class A	10	↑	↓	
▲ETMIT 231248.49S 0273742.69E						

UT433						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	322º 142º 70 NM	FL460 FL245 Class A	10	↑	↓	
▲XARMA 244331.61S 0300328.12E						
	321º 141º 25 NM	FL460 FL245 Class A	10	↑	↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲UTUPU 242925.25S 0294048.28E						
	320º 140º 43 NM	FL460 FL245 Class A	10	↑	↓	
▲DUMGO 240510.31S 0290150.64E						
	320º 139º 27 NM	FL460 FL245 Class A	10	↑	↓	
▲UTIGU 234947.99S 0283708.32E						
	319º 139º 23 NM	FL460 FL245 Class A	10	↑	↓	
▲IMPOD 233643.21S 0281607.85E						
	319º 138º 10 NM	FL460 FL245 Class A	10	↑	↓	
▲EKBAP 233100.95S 0280657.94E						
	318º 138º 32.46 NM	FL460 FL245 Class A	10	↑	↓	
▲ETMIT 231248.49S 0273742.69E						

T444						
▲DUTGI 251743.77S 0320125.26E						
	282º 50 NM	FL245 FL200 Class A	10		↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						

UT444						
▲DUTGI 251743.77S 0320125.26E						
	282º 50 NM	FL460 FL245 Class A	10		↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						

T446						
▲UNPEN 250643.41S 0320157.14E						
	270º 53 NM	FL245 FL200 Class A	10		↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	273º 30 NM	FL245 FL200 Class A	10		↓	
▲DODOS 253032.17S 0303437.62E						
	273º 77 NM	FL245 FL200 Class A	10		↓	
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						

UT446						
▲UNPEN 250643.41S 0320157.14E						
	270º 53 NM	FL460 FL245 Class A	10		↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	273º 30 NM	FL460 FL245 Class A	10		↓	
▲DODOS 253032.17S 0303437.62E						
	273º 77 NM	FL460 FL245 Class A	10		↓	
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Q4						
▲EXOBI 260252.27S 0290851.32E						
	120° 90 NM	FL245 FL200 Class A	10	↓		
▲DUKRA 262024.43S 0304711.57E						
	129° 31 NM	FL245 FL200 Class A	10	↓		
▲IXESU 263038.00S 0311959.00E						
	142° 53 NM	FL245 FL200 Class A	10	↓		
▲ALENI 265835.54S 0320937.89E						

UQ4						
▲EXOBI 260252.27S 0290851.32E						
	120° 90 NM	FL460 FL245 Class A	10	↓		
▲DUKRA 262024.43S 0304711.57E						
	129° 31 NM	FL460 FL245 Class A	10	↓		
▲IXESU 263038.00S 0311959.00E						
	142° 53 NM	FL460 FL245 Class A	10	↓		
▲ALENI 265835.54S 0320937.89E						

Q5						
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						
	144º 325º 64 NM	FL245 FL200 Class A	10	↓	↑	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲ESRET 291639.66S 0330805.46E						
	145º 326º 50 NM	FL245 FL200 Class A	10	↓	↑	
▲GEVIS 294137.49S 0335737.56E						

UQ5						
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						
	144º 325º 64 NM	FL460 FL245 Class A	10	↓	↑	
▲ESRET 291639.66S 0330805.46E						
	145º 326º 50 NM	FL460 FL245 Class A	10	↓	↑	
▲GEVIS 294137.49S 0335737.56E						

Q6						
▲IMKUP 244915.70S 0281420.65E						
	036º 63 NM	FL245 FL200 Class A	10	↓		
▲UTIGU 234947.99S 0283708.32E						
	040º 104 NM	FL245 FL200 Class A	10	↓		
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						

UQ6						
▲IMKUP 244915.70S 0281420.65E						
	036º 63 NM	FL460 FL245 Class A	10	↓		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲UTIGU 234947.99S 0283708.32E						
	040º 104 NM	FL460 FL245 Class A	10	↓		
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						

Q17						
▲GREYTOWN (GYV) VOR/DME 290728.99S 0303508.05E						
	40.6 NM	FL245 FL200 Class A	10	↓	↑	
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	138.62NM	FL245 FL200 Class A	10	↓	↑	
▲ANVED 310939.47S 0330634.63E						

UQ17						
▲GREYTOWN (GYV) VOR/DME 290728.99S 0303508.05E						
	40.6 NM	FL460 FL245 Class A	10	↓	↑	
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	138.62 NM	FL460 FL245 Class A	10	↓	↑	
▲ANVED 310939.47S 0330634.63E						

Q20		
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E		

Route designator Name of significant points Coordinates	*Route MAG Track MAG DIST*	*Upper limit Lower limit Classification*	*Lateral limits NM*	*Direction of cruising levels*		*Remarks Controlling unit Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	323º 77 NM	FL245 FL200 Class A	10		↓	
△ESVIR 280518.54S 0304943.75E						
	322º 48 NM	FL245 FL200 Class A	10		↓	
▲EVARI 274048.91S 0300315.19E						
	322º 56 NM	FL245 FL200 Class A	10		↓	
▲EPSAK 271154.98S 0290941.52E						

UQ20						
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						
	323º 77 NM	FL460 FL245 Class A	10		↓	
△ESVIR 280518.54S 0304943.75E						
	322º 48 NM	FL460 FL245 Class A	10		↓	
▲EVARI 274048.91S 0300315.19E						
	322º 56 NM	FL460 FL245 Class A	10		↓	
▲EPSAK 271154.98S 0290941.52E						

Q21						
▲ALENI 265835.54S 0320937.89E						
	144º 326º 159.2 NM	FL245 FL200 Class A	10	↓	↑	
▲EGSOX 282331.94S 0344131.38E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ21						
▲ALENI 265835.54S 0320937.89E						
	144° 326° 159.2 NM	FL460 FL245 Class A	10	↓	↑	
▲EGSOX 282331.94S 0344131.38E						

Q23						
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	219° 135 NM	FL245 FL200 Class A	10		↓	
▲NEXIT 303127.92S 0202233.37E						
	222° 118 NM	FL245 FL200 Class A	10		↓	
▲XALVA 322228.31S 0193400.22E						
	224° 28 NM	FL245 FL200 Class A	10		↓	
▲OKLOK 324829.22S 0192237.92E						
	197° 27 NM	FL245 FL200 Class A	10		↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

UQ23						
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	219° 135 NM	FL460 FL245 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲NEXIT 303127.92S 0202233.37E						
	222º 118 NM	FL460 FL245 Class A	10		↓	
▲XALVA 322228.31S 0193400.22E						
	224º 28 NM	FL460 FL245 Class A	10		↓	
▲OKLOK 324829.22S 0192237.92E						
	197º 27 NM	FL460 FL245 Class A	10		↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

Q24						
▲VASUR 253346.76S 0275335.87E						
	316º 44 NM	FL245 FL200 Class A	10		↓	
▲EVOPO 251224.18S 0271138.40E						
	312º 33 NM	FL245 FL200 Class A	10		↓	
▲DUSLI 245824.27S 0263901.66E						
	311º 36NM	FL245 FL200 Class A	10		↓	
▲ETOSA 244250.40S 0260312.98E						

UQ24						
▲VASUR 253346.76S 0275335.87E						
	316º 44 NM	FL460 FL245 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲EVOPO 251224.18S 0271138.40E						
	312º 33 NM	FL460 FL245 Class A	10		↓	
▲DUSLI 245824.27S 0263901.66E						
	311º 36NM	FL460 FL245 Class A	10		↓	
▲ETOSA 244250.40S 0260312.98E						

Q25						
▲ITROL 250207.97S 0272605.37E						
	334º 153º 57 NM	FL245 FL200 Class A	10	↑	↓	
▲RUDAS 241924.00S 0264406.00E						

UQ25						
▲ITROL 250207.97S 0272605.37E						
	334º 153º 57 NM	FL460 FL245 Class A	10	↑	↓	
▲RUDAS 241924.00S 0264406.00E						

Q28						
▲GEROX 272249.01S 0284943.11E						
	172º 40 NM	FL245 FL200 Class A	10	↓		
▲TESUN 275744.49S 0291104.73E						
	173º 106 NM	FL245 FL200 Class A	10	↓		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲GETOK 293046.69S 0300917.83E						

UQ28						
▲GEROX 272249.01S 0284943.11E						
	172º 40 NM	FL460 FL245 Class A	10	↓		
▲TESUN 275744.49S 0291104.73E						
	173º 106 NM	FL460 FL245 Class A	10	↓		
▲GETOK 293046.69S 0300917.83E						

Q29						
▲OKTAN 301758.32S 0303453.34E						
	240º 30 NM	FL245 FL200 Class A	10		↓	
▲UTIMO 304240.38S 0301506.53E						
	249º 64 NM	FL245 FL200 Class A	10		↓	
▲APLIP 312917.17S 0292312.75E						
	250º 73 NM	FL245 FL200 Class A	10		↓	
▲ETOVU 322122.29S 0282343.33E						
	258º 104 NM	FL245 FL200 Class A	10		↓	
▲XAKIM 332622.65S 0264747.86E						
	259º 37 NM	FL245 FL200 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲EPVUV 334903.31S 0261320.42E						

UQ29						
▲OKTAN 301758.32S 0303453.34E						
	240º 30 NM	FL460 FL245 Class A	10		↓	
▲UTIMO 304240.38S 0301506.53E						
	249º 64 NM	FL460 FL245 Class A	10		↓	
▲APLIP 312917.17S 0292312.75E						
	250º 73 NM	FL460 FL245 Class A	10		↓	
▲ETOVU 322122.29S 0282343.33E						
	258º 104 NM	FL460 FL245 Class A	10		↓	
▲XAKIM 332622.65S 0264747.86E						
	259º 37 NM	FL460 FL245 Class A	10		↓	
▲EPVUV 334903.31S 0261320.42E						

Q30						
▲ITKIT 332137.27S 0260950.41E						
	079º 112 NM	FL245 FL200 Class A	10	↓		
▲AXOXI 321215.60S 0275404.32E						
	075º 85 NM	FL245 FL200 Class A	10	↓		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲NIBOP 311515.73S 0290750.85E						
	073º 63 NM	FL245 FL200 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						

UQ30						
▲ITKIT 332137.27S 0260950.41E						
	079º 112 NM	FL460 FL245 Class A	10	↓		
▲AXOXI 321215.60S 0275404.32E						
	075º 85 NM	FL460 FL245 Class A	10	↓		
▲NIBOP 311515.73S 0290750.85E						
	073º 63 NM	FL460 FL245 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						

Q31						
▲ESPOG 331104.75S 0265046.66E						
	280º 36 NM	FL245 FL200 Class A	10		↓	
▲ITKIT 332137.27S 0260950.41E						
	289º 145 NM	FL245 FL200 Class A	10		↓	
▲GABGO 334003.20S 0231723.41E						
	289º 46 NM	FL245 FL200 Class A	10		↓	
▲ETGAX 334524.05S 0222220.04E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	290º 70 NM	FL245 FL200 Class A	10		↓	
▲UNRES 335243.30S 0205856.24E						
	290º 59 NM	FL245 FL200 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

UQ31						
▲ESPOG 331104.75S 0265046.66E						
	280º 36 NM	FL460 FL245 Class A	10		↓	
▲ITKIT 332137.27S 0260950.41E						
	289º 145 NM	FL460 FL245 Class A	10		↓	
▲GABGO 334003.20S 0231723.41E						
	289º 46 NM	FL460 FL245 Class A	10		↓	
▲ETGAX 334524.05S 0222220.04E						
	290º 70 NM	FL460 FL245 Class A	10		↓	
▲UNRES 335243.30S 0205856.24E						
	290º 59 NM	FL460 FL245 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

Q33		
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	231º 052º 60 NM	FL245 FL200 Class A	10	↑	↓	
△EKBOX 345446.09S 0250741.54E						
	230º 052º 133.2 NM	FL245 FL200 Class A	10	↑	↓	
▲EPMAK 365812.62S 0240531.84E						

UQ33						
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E						
	231º 052º 60 NM	FL460 FL245 Class A	10	↑	↓	
△EKBOX 345446.09S 0250741.54E						
	230º 052º 133.2 NM	FL460 FL245 Class A	10	↑	↓	
▲EPMAK 365812.62S 0240531.84E						

Q34						
▲IMSOM 335603.90S 0180031.24E						
	347º 36 NM	FL245 FL200 Class A	10		↓	
▲UVIKU 332734.47S 0173425.87E						

UQ34						
▲IMSOM 335603.90S 0180031.24E						
	347º 36 NM	FL460 FL245 Class A	10		↓	
▲UVIKU 332734.47S 0173425.87E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Q35						
▲UVIKU 332734.47S 0173425.87E						
	168º 48 NM	FL245 FL200 Class A	10	↓		
▲ASPIK 340618.32S 0180800.81E						

UQ35						
▲UVIKU 332734.47S 0173425.87E						
	168º 48 NM	FL460 FL245 Class A	10	↓		
▲ASPIK 340618.32S 0180800.81E						

Q36						
▲IMSOM 335603.90S 0180031.24E						
	283º 50 NM	FL245 FL200 Class A	10		↓	
▲APKIN 340500.64S 0170059.32E						

UQ36						
▲IMSOM 335603.90S 0180031.24E						
	283º 50 NM	FL460 FL245 Class A	10		↓	
▲APKIN 340500.64S 0170059.32E						

Q37						
▲APKIN 340500.64S 0170059.32E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	115° 56 NM	FL245 FL200 Class A	10	↓		
▲ASPIK 340618.32S 0180800.81E						

UQ37						
▲APKIN 340500.64S 0170059.32E						
	115° 56 NM	FL460 FL245 Class A	10	↓		
▲ASPIK 340618.32S 0180800.81E						

Q40						
▲NESAN 252948.88S 0280705.56E						
	355° 40 NM	FL245 FL200 Class A	10		↓	
▲EVIPI 245200.20S 0275054.35E						

UQ40						
▲NESAN 252948.88S 0280705.56E						
	355° 40 NM	FL460 FL245 Class A	10		↓	
▲EVIPI 245200.20S 0275054.35E						

Q41						
▲VASUR 253346.76S 0275335.87E						
	014° 42 NM	FL245 FL200 Class A	10	↓		
▲EVIPI 245200.20S 0275054.35E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ41						
▲VASUR 253346.76S 0275335.87E						
	014° 42 NM	FL460 FL245 Class A	10	↓		
▲EVIPI 245200.20S 0275054.35E						

Q42						
▲KODES 330840.00S 0184520.00E						
	081° 37 NM	FL245 FL200 Class A	10	↓		
▲OKLOK 324829.22S 0192237.92E						

UQ42						
▲KODES 330840.00S 0184520.00E						
	081° 37 NM	FL460 FL245 Class A	10	↓		
▲OKLOK 324829.22S 0192237.92E						

Q43						
▲KODES 330840.00S 0184520.00E						
	001° 169 NM	FL245 FL200 Class A	10	↓		
▲UTUMI 303219.28S 0172742.93E						
	358° 127 NM	FL245 FL200 Class A	10	↓		
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ43						
▲KODES 330840.00S 0184520.00E						
	001° 169 NM	FL460 FL245 Class A	10	↓		
▲UTUMI 303219.28S 0172742.93E						
	358° 127 NM	FL460 FL245 Class A	10	↓		
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						

Q44						
▲ITROL 250207.97S 0272605.37E						
	195° 41 NM	FL245 FL200 Class A	10		↓	
▲AVAGO 254309.90S 0272639.36E						

UQ44						
▲ITROL 250207.97S 0272605.37E						
	195° 41 NM	FL460 FL245 Class A	10		↓	
▲AVAGO 254309.90S 0272639.36E						

UN181						
▲TETUS 253134.40S 0253820.71E						
	113° 67 NM	FL460 FL200 Class A	10	↓		
▲PEDIL 253933.67S 0265134.05E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UN188						
▲UVIKU 332734.47S 0173425.87E						
	013° 178 NM	FL460 FL245 Class A	10	↓		
▲TEXAB 303203.80S 0165619.09E						
	011° 119 NM	FL460 FL245 Class A	10	↓		
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	002° 182° 15 NM	FL460 FL245 Class A	10	↓	↑	
▲ETMUS 281959.79S 0162705.83E						
	002° 181° 51 NM	FL460 FL245 Class A	10	↓	↑	
▲XUDAN 273041.21S 0161006.65E						

UN186						
▲KODES 330840.00S 0184520.00E						
	015° 158 NM	FL460 FL245 Class A	10	↓		
▲XAVMU 303228.70S 0181704.20E						
	013° 124 NM	FL460 FL245 Class A	10	↓		
▲DUDMO 282926.82S 0175551.82E						
	013° 15 NM	FL460 FL245 Class A	10	↓		
▲ITNOX 281434.29S 0175321.38E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	015º 15 NM	FL460 FL245 Class A	10	↓		
▲UVILA 275940.06S 0175212.23E						
	014º 29 NM	FL460 FL245 Class A	10	↓		
▲NEVAR 273048.35S 0174957.80E						

UN185						
▲KEBAT 273000.00S 0181800.00E						
	194º 23 NM	FL460 FL245 Class A	10		↓	
▲MITAR 275243.46S 0181951.86E						
	194º 15 NM	FL460 FL245 Class A	10		↓	
▲EGTAR 280738.17S 0182105.08E						
	188º 15 NM	FL460 FL245 Class A	10		↓	
▲UNGOS 282225.46S 0182410.84E						
	188º 132 NM	FL460 FL245 Class A	10		↓	
▲OKRUL 303223.92S 0185203.44E						
	191º 104 NM	FL460 FL245 Class A	10		↓	
▲GEXIP 321459.57S 0191458.08E						
	192º 34 NM	FL460 FL245 Class A	10		↓	
▲OKLOK 324829.22S 0192237.92E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Z27						
▲TAVLA 223724.00S 0281736.00E						
	204° 54.3 NM	FL245 FL200 Class A	10		↓	
▲EKBAP 233100.95S 0280657.94E						
	204° 82 NM	FL245 FL200 Class A	10		↓	
▲EVIPI 245200.20S 0275054.35E						

UZ27						
▲TAVLA 223724.00S 0281736.00E						
	204° 54.3 NM	FL460 FL245 Class A	10		↓	
▲EKBAP 233100.95S 0280657.94E						
	204° 82 NM	FL460 FL245 Class A	10		↓	
▲EVIPI 245200.20S 0275054.35E						

Z29						
▲GREYTOWN (GYV) VOR/DME 290728.99S 0303508.05E						
	349° 93 NM	FL245 FL200 Class A	10		↓	
▲BEBAS 274918.48S 0293654.69E						
	347° 44 NM	FL245 FL200 Class A	10		↓	
▲EPSAK 271154.98S 0290941.52E						
	352° 34 NM	FL245 FL200 Class A	10		↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						

UZ29						
▲GREYTOWN (GYV) VOR/DME 290728.99S 0303508.05E						
	349º 93 NM	FL460 FL245 Class A	10		↓	
▲BEBAS 274918.48S 0293654.69E						
	347º 44 NM	FL460 FL245 Class A	10		↓	
▲EPSAK 271154.98S 0290941.52E						
	352º 34 NM	FL460 FL245 Class A	10		↓	
▲STANDERTON (STV) VOR 264148.19S 0285202.86E						

Z33						
▲EPSEK 232715.00S 0313345.00E						
	110.6 NM	FL245 FL200 Class A	10		↓	
▲TIDUP 242415.07S 0295011.79E						
	10 NM	FL245 FL200 Class A	10		↓	
▲UTUPU 242925.25S 0294048.28E						
	10 NM	FL245 FL200 Class A	10		↓	
▲APMUX 243435.59S 0293124.49E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	42.9 NM	FL245 FL200 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						

UZ33						
▲EPSEK 232715.00S 0313345.00E						
	110.6 NM	FL460 FL245 Class A	10		↓	
▲TIDUP 242415.07S 0295011.79E						
	10 NM	FL460 FL245 Class A	10		↓	
▲UTUPU 242925.25S 0294048.28E						
	10 NM	FL460 FL245 Class A	10		↓	
▲APMUX 243435.59S 0293124.49E						
	42.9 NM	FL460 FL245 Class A	10		↓	
▲NESAG 245648.20S 0285103.54E						

Z36						
▲TEVAP 244430.00S 0320006.00E						
	249° 070° 42 NM	FL245 FL200 Class A	10	↑	↓	
▲DUSKO 251028.74S 0312401.70E						
	250° 070° 20 NM	FL245 FL200 Class A	10	↑	↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	235º 056º 30 NM	FL245 FL200 Class A	10	↑	↓	
▲TILIR 254652.27S 0304640.37E						
	236º 056º 8 NM	FL245 FL200 Class A	10	↑	↓	
▲OKRIP 255258.16S 0304133.91E						
	236º 056º 8 NM	FL245 FL200 Class A	10	↑	↓	
▲ESVEV 255913.02S 0303619.18E						
	236º 057º 41 NM	FL245 FL200 Class A	10	↑	↓	
▲UTSEV 263154.91S 0300837.61E						
	237º 059º 94 NM	FL245 FL200 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						
	239º 059º 15 NM	FL245 FL200 Class A	10	↑	↓	
▲IXARA 275827.43S 0285340.08E						
	264º 085º 24 NM	FL245 FL200 Class A	10	↑	↓	
▲OKDIN 280916.75S 0282923.84E						
	265º 085º 19 NM	FL245 FL200 Class A	10	↑	↓	
▲XORER 281735.22S 0281037.24E						
	265º 085º 30 NM	FL245 FL200 Class A	10	↑	↓	
▲IXEKI 283043.45S 0274036.85E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels* *Odd*	*Even*	*Remarks* *Controlling unit* *Frequency*
1	*2*	*3*	*4*	*5*		*6*
	265º 085º 17 NM	FL245 FL200 Class A	10	↑	↓	
▲EPTOT 283803.09S 0272343.09E						
	265º 086º 23 NM	FL245 FL200 Class A	10	↑	↓	
▲NIDIG 284806.79S 0270018.48E						
	270º 090º 11 NM	FL245 FL200 Class A	10	↑	↓	
▲UTONU 285216.19S 0264835.09E						
	264º 085º 30 NM	FL245 FL200 Class A	10	↑	↓	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.30E						

UZ36						
▲TEVAP 244430.00S 0320006.00E						
	249º 070º 42 NM	FL460 FL245 Class A	10	↑	↓	
▲DUSKO 251028.74S 0312401.70E						
	250º 070º 20 NM	FL460 FL245 Class A	10	↑	↓	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	235º 056º 30 NM	FL460 FL245 Class A	10	↑	↓	
▲TILIR 254652.27S 0304640.37E						
	236º 056º 8 NM	FL460 FL245 Class A	10	↑	↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲OKRIP 255258.16S 0304133.91E						
	236º 056º 8 NM	FL460 FL245 Class A	10	↑	↓	
▲ESVEV 255913.02S 0303619.18E						
	236º 057º 41 NM	FL460 FL245 Class A	10	↑	↓	
▲UTSEV 263154.91S 0300837.61E						
	237º 059º 94 NM	FL460 FL245 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						
	239º 059º 15 NM	FL460 FL245 Class A	10	↑	↓	
▲IXARA 275827.43S 0285340.08E						
	264º 085º 24 NM	FL460 FL245 Class A	10	↑	↓	
▲OKDIN 280916.75S 0282923.84E						
	265º 085º 19 NM	FL460 FL245 Class A	10	↑	↓	
▲XORER 281735.22S 0281037.24E						
	265º 085º 30 NM	FL460 FL245 Class A	10	↑	↓	
▲IXEKI 283043.45S 0274036.85E						
	265º 085º 17 NM	FL460 FL245 Class A	10	↑	↓	
▲EPTOT 283803.09S 0272343.09E						
	265º 086º 23 NM	FL460 FL245 Class A	10	↑	↓	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲NIDIG 284806.79S 0270018.48E						
	270° 090° 11 NM	FL460 FL245 Class A	10	↑	↓	
▲UTONU 285216.19S 0264835.09E						
	264° 085° 30 NM	FL460 FL245 Class A	10	↑	↓	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.30E						

Q48						
▲APDAK 263303.54S 0290252.64E						
	154° 31 NM	FL245 FL200 Class A	10	↓		
▲AVAVA 265456.32S 0292710.11E						
	164° 56 NM	FL245 FL200 Class A	10	↓		
▲EVARI 274048.91S 0300315.19E						
	165° 48 NM	FL245 FL200 Class A	10	↓		
△APKIK 281949.54S 0303433.16E						
	167° 34 NM	FL245 FL200 Class A	10	↓		
▲APMAT 284723.78S 0305659.63E						

UQ48		
▲APDAK 263303.54S 0290252.64E		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	154° 31 NM	FL460 FL245 Class A	10	↓		
▲AVAVA 265456.32S 0292710.11E						
	164° 56 NM	FL460 FL245 Class A	10	↓		
▲EVARI 274048.91S 0300315.19E						
	165° 48 NM	FL460 FL245 Class A	10	↓		
△APKIK 281949.54S 0303433.16E						
	167° 34 NM	FL460 FL245 Class A	10	↓		
▲APMAT 284723.78S 0305659.63E						

Q49						
▲OKRAV 335929.87S 0243612.57E						
	296° 111 NM	FL245 FL200 Class A	10		↓	
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	296° 50 NM	FL245 FL200 Class A	10		↓	
▲ETBUL 335956.49S 0212224.14E						
	297° 34 NM	FL245 FL200 Class A	10		↓	
▲APKAP 335918.93S 0204131.63E						
	297° 44 NM	FL245 FL200 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ49						
▲OKRAV 335929.87S 0243612.57E						
	296º 111 NM	FL460 FL245 Class A	10		↓	
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	296º 50 NM	FL460 FL245 Class A	10		↓	
▲ETBUL 335956.49S 0212224.14E						
	297º 34 NM	FL460 FL245 Class A	10		↓	
▲APKAP 335918.93S 0204131.63E						
	297º 44 NM	FL460 FL245 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

Q16						
▲EVIPI 245200.20S 0275054.35E						
	009º 99.6 NM	FL245 FL200 Class A	10	↓		
▲ETMIT 231248.49S 0273742.69E						

UQ16						
▲EVIPI 245200.20S 0275054.35E						
	009º 99.6 NM	FL460 FL245 Class A	10	↓		
▲ETMIT 231248.49S 0273742.69E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Z35						
▲ETMIT 231248.49S 0273742.69E						
	199º 109.49 NM	FL245 FL200 Class A	10		↓	
▲ITROL 250207.97S 0272605.37E						

UZ35						
▲ETMIT 231248.49S 0273742.69E						
	199º 109.49 NM	FL460 FL245 Class A	10		↓	
▲ITROL 250207.97S 0272605.37E						

Q52						
▲OKLOK 324829.22S 0192237.92E						
	072º 183 NM	FL245 FL200 Class A	10	↓		
▲APLEN 304538.77S 0220119.80E						
	069º 238 NM	FL245 FL200 Class A	10	↓		
▲GEPES 280036.99S 0251747.83E						
	068º 133 NM	FL245 FL200 Class A	10	↓		
▲ANIXU 263020.13S 0270738.66E						
	075º 25 NM	FL245 FL200 Class A	10	↓		
▲EGPOP 261653.97S 0273039.44E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UQ52						
▲OKLOK 324829.22S 0192237.92E						
	072º 183 NM	FL460 FL245 Class A	10	↓		
▲APLEN 304538.77S 0220119.80E						
	069º 238 NM	FL460 FL245 Class A	10	↓		
▲GEPES 280036.99S 0251747.83E						
	068º 133 NM	FL460 FL245 Class A	10	↓		
▲ANIXU 263020.13S 0270738.66E						
	075º 25 NM	FL460 FL245 Class A	10	↓		
▲EGPOP 261653.97S 0273039.44E						

Z30						
▲KODES 330840.00S 0184520.00E						
	049º 59 NM	FL245 FL200 Class A	10	↓		
▲GEXIP 321459.57S 0191458.08E						
	048º 113 NM	FL245 FL200 Class A	10	↓		
▲APKEK 303139.17S 0201015.32E						
	046º 139 NM	FL245 FL200 Class A	10	↓		
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UZ30						
▲KODES 330840.00S 0184520.00E						
	049º 59 NM	FL460 FL245 Class A	10	↓		
▲GEXIP 321459.57S 0191458.08E						
	048º 113 NM	FL460 FL245 Class A	10	↓		
▲APKEK 303139.17S 0201015.32E						
	046º 139 NM	FL460 FL245 Class A	10	↓		
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						

Z37						
▲UTIMO 304240.38S 0301506.53E						
	259º 65 NM	FL245 FL200 Class A	10		↓	
▲NEXUL 312137.95S 0291448.76E						
	259º 85 NM	FL245 FL200 Class A	10		↓	
▲AXOXI 321215.60S 0275404.32E						
	275º 80 NM	FL245 FL200 Class A	10		↓	
▲XARVI 324125.16S 0262636.17E						
	275º 24 NM	FL245 FL200 Class A	10		↓	
▲XOXOR 325008.37S 0255939.92E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	276° 145 NM	FL245 FL200 Class A	10		↓	
▲GABGO 334003.20S 0231723.41E						

UZ37						
▲UTIMO 304240.38S 0301506.53E						
	259° 65 NM	FL460 FL245 Class A	10		↓	
▲NEXUL 312137.95S 0291448.76E						
	259° 85 NM	FL460 FL245 Class A	10		↓	
▲AXOXI 321215.60S 0275404.32E						
	275° 80 NM	FL460 FL245 Class A	10		↓	
▲XARVI 324125.16S 0262636.17E						
	275° 24 NM	FL460 FL245 Class A	10		↓	
▲XOXOR 325008.37S 0255939.92E						
	276° 145 NM	FL460 FL245 Class A	10		↓	
▲GABGO 334003.20S 0231723.41E						

Q56						
▲XUGOL 341034.13S 0201127.59E						
	117° 60 NM	FL245 FL200 Class A	10	↓		
▲AXEMU 341211.61S 0212400.41E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	117° 49 NM	FL245 FL200 Class A	10	↓		
▲UVGUK 341259.00S 0222247.68E						
	117° 112 NM	FL245 FL200 Class A	10	↓		
▲OKSET 341301.09S 0243821.03E						

UQ56						
▲XUGOL 341034.13S 0201127.59E						
	117° 60 NM	FL460 FL245 Class A	10	↓		
▲AXEMU 341211.61S 0212400.41E						
	117° 49 NM	FL460 FL245 Class A	10	↓		
▲UVGUK 341259.00S 0222247.68E						
	117° 112 NM	FL460 FL245 Class A	10	↓		
▲OKSET 341301.09S 0243821.03E						

Q58						
▲GABGO 334003.20S 0231723.41E						
	089° 164 NM	FL245 FL200 Class A	10	↓		
▲TIKER 322402.12S 0261016.18E						
	088° 28 NM	FL245 FL200 Class A	10	↓		
▲IMROG 321042.91S 0263914.43E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	087° 68 NM	FL245 FL200 Class A	10	↓		
▲EXABU 313756.12S 0274846.39E						
	087° 15 NM	FL245 FL200 Class A	10	↓		
▲TIDAX 313034.55S 0280413.34E						
	086° 52 NM	FL245 FL200 Class A	10	↓		
▲DULRI 310459.02S 0285639.08E						
	086° 64 NM	FL245 FL200 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						

UQ58						
▲GABGO 334003.20S 0231723.41E						
	089° 164 NM	FL460 FL245 Class A	10	↓		
▲TIKER 322402.12S 0261016.18E						
	088° 28 NM	FL460 FL245 Class A	10	↓		
▲IMROG 321042.91S 0263914.43E						
	087° 68 NM	FL460 FL245 Class A	10	↓		
▲EXABU 313756.12S 0274846.39E						
	087° 15 NM	FL460 FL245 Class A	10	↓		
▲TIDAX 313034.55S 0280413.34E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	086° 52 NM	FL460 FL245 Class A	10	↓		
▲DULRI 310459.02S 0285639.08E						
	086° 64 NM	FL460 FL245 Class A	10	↓		
▲IMLID 303238.38S 0300129.69E						

Z20						
▲ETBIN 321231.24S 0274145.21E						
	036° 35 NM	FL245 FL200 Class A	10	↓		
▲EXABU 313756.12S 0274846.39E						
	036° 14 NM	FL245 FL200 Class A	10	↓		
▲DUDVU 312352.27S 0275136.50E						
	035° 23 NM	FL245 FL200 Class A	10	↓		
▲XADUP 310048.50S 0275613.14E						
	035° 30 NM	FL245 FL200 Class A	10	↓		
▲OKREV 303135.33S 0280200.66E						
	034° 144 NM	FL245 FL200 Class A	10	↓		
▲OKDIN 280916.75S 0282923.84E						
	031° 10 NM	FL245 FL200 Class A	10	↓		
▲DUVPI 275912.05S 0283117.44E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	015º 31 NM	FL245 FL200 Class A	10	↓		
▲OKTEK 272835.55S 0282748.16E						

UZ20						
▲ETBIN 321231.24S 0274145.21E						
	036º 35 NM	FL460 FL245 Class A	10	↓		
▲EXABU 313756.12S 0274846.39E						
	036º 14 NM	FL460 FL245 Class A	10	↓		
▲DUDVU 312352.27S 0275136.50E						
	035º 23 NM	FL460 FL245 Class A	10	↓		
▲XADUP 310048.50S 0275613.14E						
	035º 30 NM	FL460 FL245 Class A	10	↓		
▲OKREV 303135.33S 0280200.66E						
	034º 144 NM	FL460 FL245 Class A	10	↓		
▲OKDIN 280916.75S 0282923.84E						
	031º 10 NM	FL460 FL245 Class A	10	↓		
▲DUVPI 275912.05S 0283117.44E						
	015º 31 NM	FL460 FL245 Class A	10	↓		
▲OKTEK 272835.55S 0282748.16E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Z34						
▲ETGEB 323014.57S 0283503.45E						
	069º 72 NM	FL245 FL200 Class A	10	↓		
▲UTOXO 313621.38S 0293059.82E						
	068º 97 NM	FL245 FL200 Class A	10	↓		
▲DUKMU 302248.58S 0304500.43E						

UZ34						
▲ETGEB 323014.57S 0283503.45E						
	069º 72 NM	FL460 FL245 Class A	10	↓		
▲UTOXO 313621.38S 0293059.82E						
	068º 97 NM	FL460 FL245 Class A	10	↓		
▲DUKMU 302248.58S 0304500.43E						

Z31						
▲XAGEN 260344.92S 0271904.08E						
	246º 32 NM	FL245 FL200 Class A	10		↓	
▲XANRO 262507.09S 0265233.55E						
	245º 22 NM	FL245 FL200 Class A	10		↓	
▲EXOKU 264010.40S 0263440.15E						
	245º 117 NM	FL245 FL200 Class A	10		↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲IMSIR 275945.73S 0245801.19E						
	245º 232 NM	FL245 FL200 Class A	10		↓	
▲NESUS 304028.10S 0214554.26E						
	250º 152 NM	FL245 FL200 Class A	10		↓	
▲XALVA 322228.31S 0193400.22E						

UZ31						
▲XAGEN 260344.92S 0271904.08E						
	246º 32 NM	FL460 FL245 Class A	10		↓	
▲XANRO 262507.09S 0265233.55E						
	245º 22 NM	FL460 FL245 Class A	10		↓	
▲EXOKU 264010.40S 0263440.15E						
	245º 117 NM	FL460 FL245 Class A	10		↓	
▲IMSIR 275945.73S 0245801.19E						
	245º 232 NM	FL460 FL245 Class A	10		↓	
▲NESUS 304028.10S 0214554.26E						
	250º 152 NM	FL460 FL245 Class A	10		↓	
▲XALVA 322228.31S 0193400.22E						

Q51		
▲ESRUK 335810.48S 0194826.84E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	268° 11 NM	FL245 FL200 Class A	10		↓	
▲GETEN 340326.38S 0193611.70E						

UQ51						
▲ESRUK 335810.48S 0194826.84E						
	268° 11 NM	FL460 FL245 Class A	10		↓	
▲GETEN 340326.38S 0193611.70E						

UL211F						
▲APKIN 340500.64S 0170059.32E						
	257° 079° 122 NM	FL460 FL200 Class A	10	↑	↓	
▲ITLIK 351600.00S 0145957.00E						
	271° 094° 261 NM	FL460 FL200 Class A	10	↑	↓	
▲ANTES 365100.00S 0100000.00E						
	274° 100° 494 NM	FL460 FL200 Class A	10	↑	↓	
▲EKBED 3391300.00S 0000000.00W						
	282° 108° 467 NM	FL460 FL200 Class A	10	↑	↓	
▲MUNES 402000.00S 0100000.00W						

UL224		
▲APKIN 340500.64S 0170059.32E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	289° 111° 101 NM	FL460 FL200 Class A	10	↑	↓	
▲ITMEK 3341200.00S 00150000.00E						
	290° 113° 249 NM	FL460 FL200 Class A	10	↑	↓	
▲ETULA 342100.00S 0100000.00E						
	293° 119° 498 NM	FL460 FL200 Class A	10	↑	↓	
▲GERAM 340300.00S 0000000.00W						
	299° 125° 506 NM	FL460 FL200 Class A	10	↑	↓	
▲ITGIV 325600.00S 0100000.00W						

UL435						
▲O R TAMBO INTERNATIONAL AIRPORT (JSV) VOR/DME 260925.63S 0281351.70E						
	341° 161° 36 NM	FL460 FL200 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						
	318° 137° 77 NM	FL460 FL200 Class A	10	↑	↓	
▲NESEK 250154.00S 0263700.00E						FAJA/FBGR FIR BDRY

UT915						
▲ITMIL 290151.05S 0314838.40E						
	011° 189° 105 NM	FL460 FL290 Class A	10	↓	↑	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲GERAX 271857.48S 0312310.91E						
	018º 197º 48 NM	FL460 FL290 Class A	10	↓	↑	
▲IXESU 263038.00S 0311959.00E						
	010º 189º 34 NM	FL460 FL290 Class A	10	↓	↑	
▲DULPU 255723.71S 0311325.26E						
	009º 189º 6 NM	FL460 FL290 Class A	10	↓	↑	
▲NEXIS 255115.01S 0311212.83E						
	009º 188º 29 NM	FL460 FL290 Class A	10	↓	↑	
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	020º 197º 89 NM	FL460 FL245 Class A	10	↓	↑	
▲PHALABORWA (PHV) VOR 235402.73S 0310907.09E						
	011º 189º 95 NM	FL460 FL245 Class A	10	↓	↑	
▲GESIL 221900.00S 0310000.00E						

UT916						
▲UDLUM 251622.00E 0232313.09E						
	221º 044º 147 NM	FL 460 FL 290 Class A	10	↑	↓	
▲AVUSA 272955.10S 0221330.09E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	242º 063º 75 NM	FL 460 FL 290 Class A	10	↑	↓	
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						

Q59						
▲ESPUV 254945.36S 0244255.80E						
	236º 059º 143 NM	FL245 FL200 Class A	10	↑	↓	
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						

UQ59						
▲ESPUV 254945.36S 0244255.80E						
	236º 059º 143 NM	FL460 FL245 Class A	10	↑	↓	
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						

Q60						
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						
	237º 220 NM	FL245 FL200 Class A	10		↓	
▲NEXIT 303127.92S 0202233.37E						

UQ60		
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	237° 220 NM	FL460 FL245 Class A	10		↓	
▲NEXIT 303127.92S 0202233.37E						

Q61						
▲APKEK 303139.17S 0201015.32E						
	063° 228 NM	FL245 FL200 Class A	10	↓		
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						

UQ61						
▲APKEK 303139.17S 0201015.32E						
	063° 228 NM	FL460 FL245 Class A	10	↓		
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						

Z39						
▲TETAN 333948.00S 0192458.00E						
	089° 54 NM	FL245 FL200 Class A	10	↓		
▲TEXIS 331556.01S 0202256.73E						
	075° 193 NM	FL245 FL200 Class A	10	↓		
▲EGRUP 311005.95S 0231539.78E						
	073° 133 NM	FL245 FL200 Class A	10	↓		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲IBTEK 294304.04S 0251247.52E						
	080° 68 NM	FL245 FL200 Class A	10	↓		
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						

UZ39						
▲TETAN 333948.00S 0192458.00E						
	089° 54 NM	FL460 FL245 Class A	10	↓		
▲TEXIS 331556.01S 0202256.73E						
	075° 193 NM	FL460 FL245 Class A	10	↓		
▲EGRUP 311005.95S 0231539.78E						
	073° 133 NM	FL460 FL245 Class A	10	↓		
▲IBTEK 294304.04S 0251247.52E						
	080° 68 NM	FL460 FL245 Class A	10	↓		
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						

Z40						
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						
	249° 113 NM	FL245 FL200 Class A	10		↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						
	251° 80 NM	FL245 FL200 Class A	10		↓	
▲TIKIL 311522.20S 0233202.10E						
	252° 213 NM	FL245 FL200 Class A	10		↓	
▲UVDEX 333359.30S 0202034.70E						
	253° 36 NM	FL245 FL200 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

UZ40						
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						
	249° 113 NM	FL460 FL245 Class A	10		↓	
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						
	251° 80 NM	FL460 FL245 Class A	10		↓	
▲TIKIL 311522.20S 0233202.10E						
	252° 213 NM	FL460 FL245 Class A	10		↓	
▲UVDEX 333359.30S 0202034.70E						
	253° 36 NM	FL460 FL245 Class A	10		↓	
▲ESRUK 335810.48S 0194826.84E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
Z41						
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	113° 129 NM	FL245 FL200 Class A	10	↓		
▲UTERI 283214.47S 0234208.26E						
	113° 79 NM	FL245 FL200 Class A	10	↓		
▲EVESI 283554.70S 0251112.70E						
	93° 46 NM	FL245 FL200 Class A	10	↓		
▲UNRAG 282149.12S 0260109.16E						

UZ41						
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	113° 129 NM	FL460 FL245 Class A	10	↓		
▲UTERI 283214.47S 0234208.26E						
	113° 79 NM	FL460 FL245 Class A	10	↓		
▲EVESI 283554.70S 0251112.70E						
	93° 46 NM	FL460 FL245 Class A	10	↓		
▲UNRAG 282149.12S 0260109.16E						

Z38		
▲DUGPA 261329.04S 0222534.83E		

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	205º 026º 77 NM	FL245 FL200 Class A	10	↑	↓	
▲AVUSA 272955.10S 0221330.09E						

UZ38						
▲DUGPA 261329.04S 0222534.83E						
	205º 026º 77 NM	FL460 FL245 Class A	10	↑	↓	
▲AVUSA 272955.10S 0221330.09E						

UT394						
▲AVUSA 272955.10S 0221330.09E						
	302º 122º 67 NM	FL460 FL245 Class A	10	↑	↓	FAJA ACC WEST 118.55 MHz
▲UTANI 271346.73S 0210034.80E						

UT396						
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						
	349º 168º 47 NM	FL460 FL245 Class A	10	↑	↓	FAJA ACC WEST 118.55 MHz
▲UDLON 274350.57S 0204847.32E						
	348º 168º 16 NM	FL460 FL245 Class A	10	↑	↓	FAJA ACC WEST 118.55 MHz
▲EGNOR 272955.15S 0203940.57E						

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
T949						
▲TONKA 260724.00S 0320618.00E						FQBE FIR/ FDMS TMA BDRY
	253º 074º 26 NM	FL245 FL200 Class A	10	↑	↓	
▲SIKHUPHE (VSK) VOR 262245.90S 0314302.70E						FDMS TMA
	269º 089º 22 NM	FL245 FL200 Class A	10	↑	↓	
▲IXESU 263038.00S 0311959E						
	253º 073º 11 NM	FL245 FL200 Class A	10	↑	↓	
▲MUVAL 263728.00S 0310957.00E						
	268º 089º 22 NM	FL245 FL200 Class A	10	↑	↓	
▲APNON 264537.86S 0304652.79E						FAJA FIR/FDMS TMA BDRY
	256º 078º 110 NM	FL245 FL200 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						

UT949						
▲TONKA 260724.00S 0320618.00E						FQBE FIR/ FDMS TMA BDRY
	253º 074º 26 NM	FL460 FL245 Class A	10	↑	↓	
▲SIKHUPHE (VSK) VOR 262245.90S 0314302.70E						FDMS TMA
	269º 089º 22 NM	FL460 FL245 Class A	10	↑	↓	

Route designator *RCP Type* *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲IXESU 263038.00S 0311959E						
	253º 073º 11 NM	FL460 FL245 Class A	10	↑	↓	
▲MUVAL 263728.00S 0310957.00E						
	268º 089º 22 NM	FL460 FL245 Class A	10	↑	↓	
▲APNON 264537.86S 0304652.79E						FAJA FIR/FDMS TMA BDRY
	256º 078º 110 NM	FL460 FL245 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						

Q62						
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	185º 121 NM	FL245 FL200 Class A	10		↓	
▲UVBUD 303208.64S 0170448.56E						
	187º 220 NM	FL245 FL200 Class A	10		↓	
▲ASPIK 340618.32S 0180800.81E						

UQ62						
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	185º 121 NM	FL460 FL245 Class A	10		↓	
▲UVBUD 303208.64S 0170448.56E						

	187° 220 NM	FL460 FL245 Class A	10		↓	
▲ASPIK 340618.32S 0180800.81E						

Q63						
▲APMAT 284723.78S 0305659.63E						
	236° 36 NM	FL245 FL200 Class A	10		↑	
▲APGAP 281711.47S 0311946.86E						
	026° 204° 58 NM	FL245 FL200 Class A	10	↓	↑	
▲GERAX 271857.48S 0312310.91E						

UQ63						
▲APMAT 284723.78S 0305659.63E						
	236° 36 NM	FL460 FL245 Class A	10		↑	
▲APGAP 281711.47S 0311946.86E						
	026° 204° 58 NM	FL460 FL245 Class A	10	↓	↑	
▲GERAX 271857.48S 0312310.91E						